

Application: To examine the bivariate plot of two variables – usually to determine if there is a linear relationship before running a Pearson’s correlation.

Creating a Scatterplot

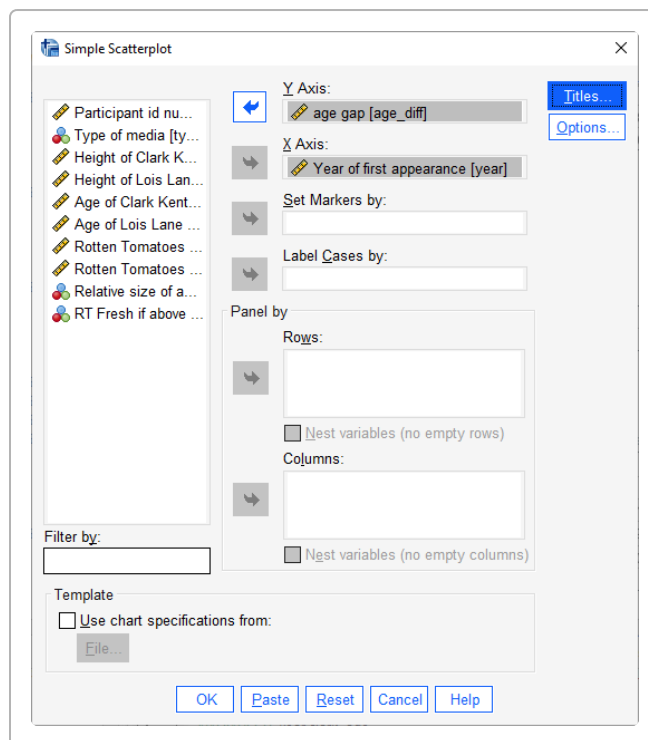
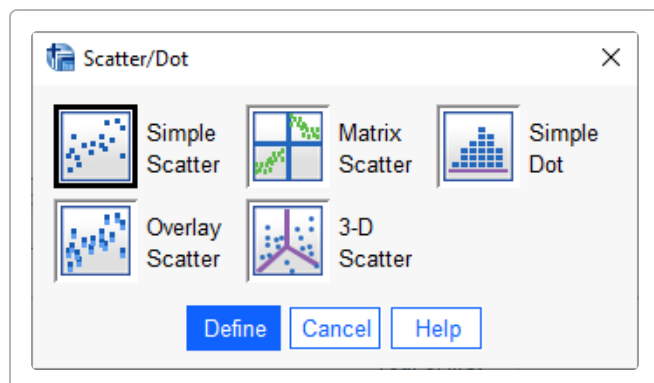
Graph → Legacy Dialogs → Scatter/Dot

- Highlight “Simple Scatter” and click “Define”
- Highlight the criterion variable and use the arrow to move into the “Y Axis” window
- Highlight the predictor variable and use the arrow to move into the “X Axis” window

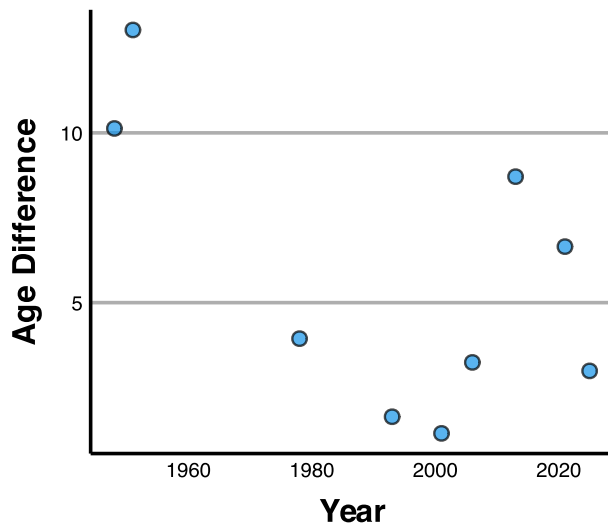
SPSS Syntax

```
GRAPH
  /SCATTERPLOT(BIVAR)=year WITH age_diff.
```

① predictor (x-axis) WITH criterion (y-axis)



SPSS Output



Inspection of the graph suggests that there is a generally linear trend to the data, with no obvious curvilinearity.

So, we would be comfortable analyzing these data using a Pearson's correlation.

The correlation coefficient ($r = -0.584$) indicates a negative relationship between year and age difference — as year increases, the age difference tends to decrease.